AI-Based Negotiation Style Training

Max Balle, Marlene Meyer, and Mareike Schoop(✉)

University of Hohenheim, Schwerzstrasse 40, 70599 Stuttgart, Germany
{max.balle,marlene.meyer,schoop}@uni-hohenheim.de

Abstract. Knowing one's own negotiation style and recognising the partner's negotiation style are fundamental prerequisite to negotiate effectively. Since novice negotiators often fail to state these styles correctly, a dedicated training to support the identification of negotiation styles is required. In this paper, an AI-based training approach is presented. As a first step, software agents acting as training partners have to be designed. We implemented a genetic algorithm to shape the agents' behaviour. As a first implication, the designed agents act according to their intended negotiation styles without the need for extensive datasets of previous negotiations to specify those styles.

Keywords: Negotiation Styles · E-Training · Digital Negotiation · Agents · Genetic Algorithm

1 Introduction

Digital negotiations are present in everyday life [24]. They provide dedicated negotiation support through the use of information technology by means of negotiation support systems (NSSs). The support functionalities comprise communication support, decision support, document management, and conflict management [39].

Digital negotiations require system skills as well as negotiation skills. Therefore, dedicated training for both is required [25, 26]. If effective and efficient digital negotiations are the goal, the training has to be conducted in a digital form, e.g. among human negotiators using an NSS [42] or in a human-machine interaction via NSSs where the machine part is performed by intelligent software agent. The latter has the advantage of not relying on the availability and skills of other human negotiators and of not being dependent on moods, expertise, competencies, motivation and engagement of those [9, 22]. Negotiation agents are software that makes decisions in negotiations on behalf of a human principal, and thus, automate at least one negotiation activity, such as conducting in negotiations [16, 21]. Negotiation agents have been used extensively in the early 2000s. However, agents up to now do not consider individual negotiation styles in detail rather than providing behaviour orientations (cooperation vs. competition) nor do they apply different styles in training human negotiators [1, 5, 12, 31]. However, agents have the potential of displaying several styles and thus enabling novice negotiators to experience such behaviour, to learn how to react, and, most importantly, to recognise their own negotiation style(s). It was shown that novices have problems with all of these goals

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[28]. This paper thus designs negotiation agents with an AI-based approach for teaching various negotiation styles to (novice) negotiators focusing on the decision-making process in a first step. An additional benefit of our AI approach with a decision-making focus is that it does not require the large amount of training data usually called for in AI applications.

Thus, the aim of this research contribution is to 1) design AI-agents for the decision-making process in negotiations without the need of extensive learning data, and 2) examine to what extent those agents can imitate the negotiation styles.

2 Methodological Approach

The methodology is based on a design science research (DSR) approach [13] (cf. Fig. 1).

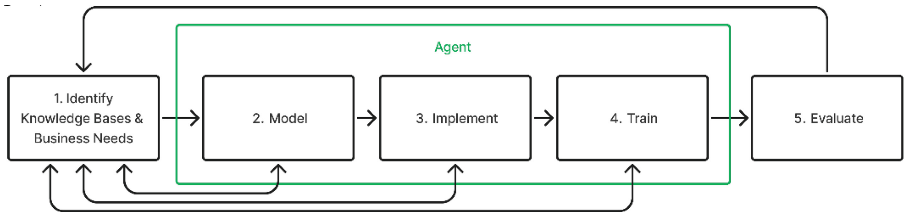

Fig. 1. Methodical Approach

First, the knowledge base and the business needs are identified. As negotiators have to be trained and since such training through an agent can be more beneficial than with a human partner [11], we examined the literature (i.e. knowledge base) for negotiation agents. Since our findings revealed that those agents do not consider negotiation theories, especially regarding negotiation styles, the aim of our research contribution (i.e. business needs) is to design and implement agents considering negotiation styles for decision-making by using an AI-based approach.

For each of the following phases (phase 2–5), the previous step is used as a basis and is adapted with new knowledge from the knowledge base if the provided information from the previous step is not sufficient.

The second phase includes the modelling of the artefact by identifying requirements of the agents, and designing a web-based system to enable human negotiators (later called participants) to negotiate with those agents. The requirements for the agents are identified by conducting a systematic literature search, which is based on an adapted four-step process as outlined by [40]. The systematic literature search has considered peer-reviewed articles in English only, with the publication date ranging from 2017 to the present. The search strings encompass agents, training, and negotiation.

In order to implement the agents in phase 3, an AI model has to be chosen. As neither a fitting pre-trained model nor a sufficiently large dataset of previous negotiations are available, these conditions have to be considered in the selection of the appropriate AI model for our agents.

In the training phase, the implemented agents are trained through conducting multiple negotiations with other agents. A detailed description of the training is presented in Sect. 5.

In the last phase, an experiment is conducted with twenty-seven participants who negotiate with the agent and fill in a survey.

3 Theoretical Background

3.1 Negotiation Styles

Negotiation styles are orientations how someone can behave in a negotiation. The orientations are based on how strong the concerns about one's own goals and the relationship to the partner are pronounced. Regarding this, five types of negotiation styles are derived: accommodating when the relationship to the partner is most important and is shown by yielding to the partner's preferences; competing when the own outcome is the most important; compromising when both interests are of concern and a compromise is attempted, e.g., by splitting the differences; collaborating when a win-win result satisfying both parties is tried to be achieved, and avoiding when neither the relationship nor the own outcome are relevant; [23].

3.2 Artificial Intelligence

There are two dimensions of AI to consider 1) measurement of intelligence in AI (is the intelligence of AI measured in comparison to human intellect or with respect to rationality?) [33], and 2) quality type of intelligence (do we search for intelligence as an internal quality or externally in the form of intelligent behaviour?) [27, 33]. Rationality refers to choosing the right action with respect to a certain goal and considering resource limitations [14]. Based on these two dimensions, four different perspectives on the topic are derived, namely acting humanly, thinking humanly, thinking rationally, and acting rationally [33] (cf. Fig. 2).

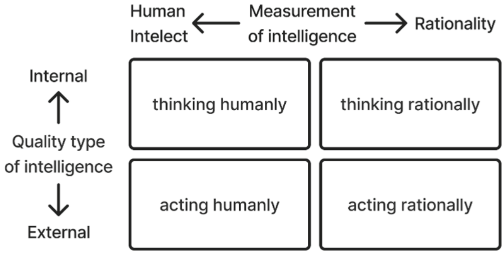

Fig. 2. Perspectives on AI (adapted from [33])

Amongst the current AI research, the perspective *acting rationally* is the most dominant approach [33]. One of these branches of AI is Machine Learning (ML). As a scientific field, ML provides computers with the ability to learn without explicit programming [35]. ML is programming computers "to learn from experience [to] eliminate the need for much [...] programming effort" [35]. A precise definition of ML programs is given by Mitchell: "A computer program is said to learn from experience E with respect to some class of tasks T and performance measure P, if its performance at tasks in T, as measured by P, improves with experience E" [30].

The three main types of ML are supervised, unsupervised, and reinforcement learning. Supervised learning trains the AI with labelled data – pairs of data with input and expected corresponding output – while unsupervised learning uses unlabelled data, i.e., without receiving explicit feedback [34]. Reinforcement learning allows the agent to learn from its actions by receiving feedback in form of rewards or punishments [34].

There are also emergent models of learning, such as the genetic algorithm, which is inspired by evolution [27]. Genetic algorithms (GAs) are characterised by evolution through mating and mutation [15]. GAs work by first initialising a population of potential solutions, then the population is repeatedly evaluated, reproduced, and replaced. The evaluation is based on evaluating the fitness of the individuals from the population for the respective task. During the reproduction, descendants are generated from the population by using genetic operators. Finally, the descendants are used to replace a part – mostly the unfit individuals – of the previous population [27]. Multiple strategies exists for the selection process and for the genetic operators recombination and mutation [20, 29].

Deep Learning is a specific type of machine learning, which is characterised by neural networks with many layers. Artificial Neural Networks (ANNs) are inspired by neurons [27] and can represent complex non-linear functions with computational circuits [34]. ANNs are called feedforward neural networks, if the output of each unit in the network only connects to the next layers. ANNs are defined as recurrent neural networks (RNNs), if units can cycle their outputs back into previous layers [34]. This cycling results in RNNs being able to memorise, thus, making them particularly useful with sequential data [33]. Gated Recurrent Units (GRU) are a type of RNNs proposed by Cho et al. in 2014 [4] offering improved memory length and reduced the risk of vanishing or exploding gradients [6].

4 Designed Artefact

In this chapter, the artefact for negotiation style training with AI-based agents is designed including the derived requirements for the artefact, a detailed description of the client-server-architecture of the system, and the designed agents.

4.1 Requirements

The systematic literature review was conducted using search strings including the terms "agents", "training", and "negotiation" in order to derive requirements for the intended agents. The review yielded six requirements, which are listed in descending order according to the frequency with which they were identified in the literature search. The first requirement is generally the most relevant to implement for our purpose. However, depending on the focus of the application of the agents, the order might change.

1. *Realistic*: Most importantly, agents have to behave realistically. This is supposed to ensure transferability of the learning to real-world human-to-human negotiation. [7, 41, 43].
2. *First-order Theory of Mind (ToM)*: This is the second most common requirement. To negotiate realistically and integratively, the agents must be capable of reasoning over beliefs and intentions the other side holds – especially regarding the utility values. [3, 17, 36, 41].
3. *Feedback*: For trainees, appropriate feedback is of high importance when learning the skills that comprise negotiation [7, 18, 44].
4. *Share information*: Agents should be able to implicitly and/or explicitly share information, mostly about their preferences. The negotiation partners rely on this information to perform ToM themselves and overcome the fixed-pie-bias [17, 36].
5. *No knowledge about the partner's utility*: Just like in most real-life negotiations, agents should have no prior knowledge of their partner's utility values. This requirement is also the reason agents have to perform ToM and engage in information sharing. [19, 32].
6. *Use history of exchange*: Finally, the agents have to be able to factor all prior exchanged messages into their decision on the next move. Again, this is necessary for realistic behaviour and to perform ToM [37, 43].

Regarding the quality type of intelligence (cf. Fig. 2), all of the requirements contain the external view on how the agent directly interacts with participants. Requirements 2, 5 and 6 also consider the internal view on the agent's decision-making process.

4.2 Client-Server Architecture of the Web-Based System

The focus of this work is on designing agents as realistic negotiation partners (cf. Sect. 4.1) that support negotiation training, hence a corresponding client-server-architecture for negotiation training is designed (cf. Fig. 3).

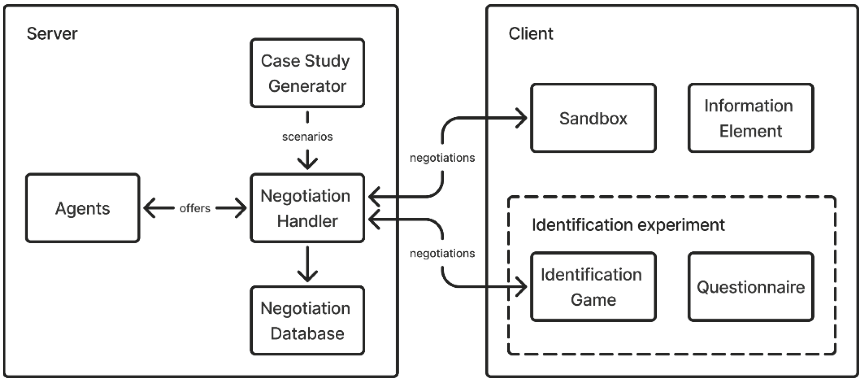

Fig. 3. Architecture

The client provides the user interface to enable the participants to negotiate with the agents either in a sandbox or during an experiment. Further, the interface provides fundamentals of negotiation theory and the user interface to ensure a uniform understanding of terms. The server contains the agents, a case study generator, and the negotiation handler with a database. The negotiation handler manages the initialisation of a new negotiation by generating a case study, and all states of the ongoing negotiations, i.e., all (counter-)offers between the negotiators (agent and participant) are managed though the negotiation handler. The case study generator and the agents are further specified in the following chapters.

4.3 Case Study Generator

The usage of a case study generator which produces new collaborative case studies with preference models for each negotiation ensures that the participants do not have a learning effect through repetitive case studies when negotiating multiple times. All case studies generated for this work contain five issues and five different options for each issue.

The relative importance of the issues for each preference model are generated independently of the partner's preference model by iteratively expanding the number of issues, starting with a set containing one issue with a relative importance of 100%. Then, the greatest value is split into two randomly sized parts with a centred triangular distribution. The calculation stops when the set contains the defined number of issues for the preference model and the issues get shuffled. The aim of this calculation is to reduce the likelihood of irrelevant issues (with a relative importance that is too low) compared to models generated by, e.g. simply splitting 100% at n-1 points.

For each issue, the preferences of each issue selection are generated considering the partner's preference model to generate actual disputes in the case study. For each issue in each preference model, the preferred option (100%) is selected at random. For both preference models, a triangular distribution is used, but one is left- sided and the other right-sided. The remaining issue selections are filled in based on the distance to the preferred option of the issue and altered by Gaussian noise.

To ensure a generated case study fulfils the requirements, their usage is verified through the availability of a fair outcome and the analysis of the pareto frontier, i.e., a pareto-optimal outcome exists with a contract imbalance of $<= 20\%$, and the pareto frontier is curved enough (extremes must be at least 0.5 apart and the area between the pareto frontier and a line connecting the extremes must be above 0.04 on the pareto graph). Thus, only case studies fulfilling the requirement are applied; otherwise, the generation process restarts.

4.4 Designed AI-Based Agents

When choosing an AI-method to be applied as a negotiation partner, the negotiation domain and the limited availability of suitable training data have to be considered besides further requirements (cf. Sect. 4.1). Characteristics of negotiations, such as its sequential nature and the dependencies on the negotiation partner regarding the outcome of the negotiation, make negotiation a complex domain [2, 38]. Based on the previous specified challenges of negotiating with a realistic agent, we use a Deep Learning approach. Following the ML definition by [30], for each negotiation style S we define the task T as "Negotiate like a human with S" and measure performance P with a function mapping negotiation result and S to a fitness score. To provide the experience E, negotiation simulations between the agents are conducted.

Deep Learning is realised by applying a Genetic Algorithm to RNNs. This utilises the abilities of RNNs to model complex non-linear functions and to remember past input [34] and combines it with GAs easy parallelisation and balance of exploration and exploitation [15, 20, 29].

The recurrent layers of the model are GRU-based and supplemented with dense, non-recurrent layers (cf. Fig. 4).

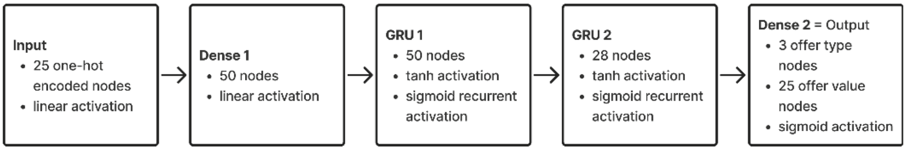

Fig. 4. RNN layers

There are fifty inputs for each new offer, two for each option in each issue: the utility of this option as the product of importance and preference, and in One-Hot-encoding, whether it is chosen. One-Hot-encoding specifies that for every issue, only the chosen option is encoded as "1", the others as "0" [10]. The model output consists of 28 nodes, a group of three for each possible message types (counteroffer/accept/reject) and five groups of five for the issues. In each group, the option corresponding to the node with the highest output is the choice of the model for its new offer.

Previous offers influence the outcome via the recurrence. Using two GRU layers allows for a recurrence along the reduction from 50 to 28 hidden units.

The performance P of an agent in a negotiation is measured with a fitness function. The fitness function varies for each negotiation style and calculates a fitness score for a negotiation outcome by calculating the weighed sum of these five factors (cf. Table 1): own utility, partner's utility, joint utility, fairness, and time. The negotiator's own utility and the partner's utility are directly based on the dimensions of the negotiation style (own concern vs. partner's concern). Joint utility is defined to measure the extent of collaborating by adding up own utility and partner's utility. Fairness specifies the contract imbalance between the own utility and the partner's utility (absolute difference). Time is considered by the additive inverse of the normalised number of exchanged messages and reflects how urgent ending or continuing the negotiation is to the agent. The time factor can change during the training progress to reduce indecisiveness and encourage agents to take more or less time while exploring solutions.

As accommodating agents prioritise their partner's utility, they disregard their own utility, joint utility, and fairness. Competing agents, as opposite to accommodating agents, prioritise their own utility and disregard their partner's utility, joint utility, and fairness. Avoiding agents try to withdraw from the negotiation by concluding as fast as possible with little care for the outcome. Collaborating agents try to completely fulfil their own and their partners' goals simultaneously, aiming for an outcome of high joint utility and fairness. Compromising agents are aiming for a partial fulfilment of their own and the partner's goals with a more or less fair outcome.

Table 1. Fitness weights

	Time	Own utility	Partner utility	Joint utility	Fairness
Accommodating	0.25/−0.025/0	0	1	0	0
Collaborating	0.1/−0.01/0	1	1	1	1
Compromising	0.75/−0.075/0	0.25	0.25	0.5	1
Avoiding	1	0	0	0	0.25
Competing	0.25/−0.025/0	1	0	0	−1

5 Training of the Agents

The designed AI-based agents have to learn how to act according to their negotiation styles through training. As a training algorithm we developed a genetic algorithm inspired by [15, 20, 29].

Due to different characteristics of each negotiation style, a different population is considered for each negotiation style in the training algorithm. Initially, there are 100 individuals per population – rising to 500 in late generations – and every agent is initialised with random weights. The genome of each individual consists of the weight matrices of its RNN. This makes this approach a real-coded GA [20]. For each generation, four consecutive steps of the GA – Negotiation simulation, selection, recombination, and mutation – are executed (cf. Fig. 5).

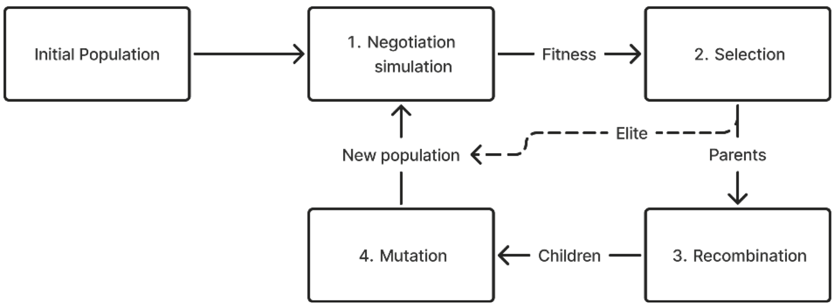

Fig. 5. Steps of the GA

After the population is initialised, the fitness of every agent is determined by a negotiation simulation (step 1). To conduct one turn of the negotiation simulation, a case study has to be used (generated by the case study generator) and the agents have to negotiate with each other. More precisely, each agent negotiates twice with one random agent of each negotiation style (in total 10 negotiations per turn) to allow switching roles and compensating unfairness in a case study. The negotiation length is limited to the exchange of 100 messages each during the negotiation to ensure termination even with indecisive models. In early generations, the negotiation simulation of one generation consists of one turn. Later generations have up to three turns to achieve higher accuracy of the estimation. The outcomes of all negotiations of an agent are judged with the fitness function under the agent's negotiation style (cf. Sect. 4.4) and the fitness scores consolidated into this agent's fitness.

Once every agent has a determined fitness, the selection (step 2) is conducted. To do so, a fixed part of the population dies without reproduction. Some approaches randomise the selection with odds favouring the fit individuals [20]. We apply a simple cut-off of the unfit 50% since step 1 already introduced chances of survival for weaker individuals when giving them easy pairings. The remaining individuals reproduce. To ensure fit individuals produce more descendants, this GA takes a staggered approach where subgroup reproduce amongst themselves; first the top 10%; followed by the top 30% and finally the top 50% (proportions taken from early trials). An elitist selection is applied to the descendants as the best individuals (top 10%) directly survive into the next generation without modification [29]. This is different to evolution in nature, but inspired by other GA approaches where the best individual survives. The benefit of elitist selection lies in the prevention of loss of development [29]. Because of luck involved in determining the fitness during step 1, more elite individuals survive in our GA, which slows down the evolution, but increases the likelihood of keeping the best agent. In each generation the same number of agents die as new agents are introduced, which keeps the size of the population consistent.

Reproduction happens in pairs of two, i.e., two parent agents combine to produce two children. During the recombination (or crossover) each of the different RNNs weight matrices of the parents' genomes get extracted and split. The split happens at a random point along on a random dimension of the matrix. Corresponding matrices of the two parents split at the same point. The weight matrices of both children are then the different combinations of the parts of the corresponding parent's matrices. This is a special case of multi-point crossover with regards to the whole genome [8] and is illustrated in Fig. 6.

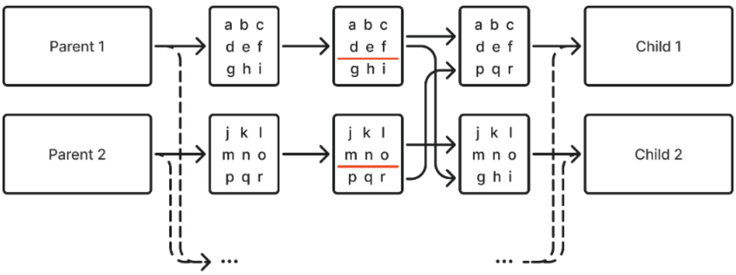

Fig. 6. Exemplary recombination of one matrix of the genome

Each child is then individually mutated, which is realised by Gaussian noise to each weight, and thus, increases the variance in the population.

When the steps of the GA terminate, the fittest individual of each of the five populations corresponding to the negotiation styles is chosen as the final model for its style by repeating step 1 of Fig. 5 with the final population to increase the likelihood of choosing the best model.

6 Results of the Experiment

In order to validate the generated agent, the defined client-server-architecture was implemented and an experiment was conducted. This chapter includes the results of the conducted experiment 1) by identifying the negotiation styles of the agents and 2) by evaluating the usefulness of the agents.

There were 27 participants in the experiment (12 female, 15 male). 21 of the participants are between 18 and 24 years old; 3 participants are between 25 and 34 years; one participant is between 45 and 54 years; and two participants are between 55 and 64 years old. 21 participants have an upper secondary education, the remaining participants have a bachelor's degree or above. 13 participants disclosed to having no prior negotiation experience; 11 participants claimed to have some experience, and three participants reported to be (highly) experienced.

In the first part of the experiment, the participants were asked to negotiate and then indicate the negotiation style they associated with their negotiation partner (aka implemented agent). The participants were asked to fill in a short survey to share some information about their demographic data and their negotiation experience. Further, they received all relevant information about the setting of the negotiation including the case study. Every case study used during the experiment was produced by our case study generator and only used once.

In each negotiation they conducted, the participants were paired with an agent of a specific negotiation style which was unknown to the participants. After completion of the negotiation, the participants were asked to indicate the negotiation style of their partner of the current negotiation. The participants were allowed to conclude as many negotiations as they like. However, none negotiated with more than two agents with the same style.

This led to 52 data points in total. 14 times the participants were able to state the negotiation style of their partner correctly. The results are differentiated between those participants with no experience (No Exp.) and those with some or more experience (Exp.) (cf. Table 2). Six participants (No Exp. 3 of 6 participants; Exp. 3 of 5 participants) identified the competing style correctly. None identified the collaborating style (No Exp. 4 participants, Exp. 2 participants) and the compromising style (No Exp. 10 participants, Exp. 2 participants) correctly. Avoiding was identified correctly five times (No Exp. 1 of 6 participants, Exp. 4 of 6 participants). Three participants identified the accommodating style correctly (No Exp. 2 of 6 participants, Exp. 1 of 4 participants).

After the participants concluded the negotiations, they were asked to complete a post-negotiation survey by rating statements on a 7-point Likert scale regarding the usefulness of the system and how realistic the agents proceeded (cf. Table 3). The conducted system

Table 2. Style Identification Differentiated by the Participants based on their Experience

Judgments		Agent				
		Competing	Collaborating	Compromising	Avoiding	Accommodating
Competing	No Exp	3	0	0	0	2
	Exp	3	0	0	0	2
Collaborating	No Exp	1	0	3	0	2
	Exp	0	0	0	0	1
Compromising	No Exp	1	0	0	2	0
	Exp	2	0	0	1	1
Avoiding	No Exp	0	2	5	1	0
	Exp	0	1	1	4	0
Accommodating	No Exp	1	2	2	3	2
	Exp	0	1	1	1	1
Total	No Exp	6	4	10	6	6
	Exp	5	2	2	6	4

Table 3. Likert statements and reactions

Statement	Mean	SD
This tool is helpful when learning about different negotiation styles	5.48	1.32
This tool is helpful to train the identification of the negotiation opponents' styles	5.38	1.20
The AI models behaved realistic	4.10	1.55
The AI models seem to work with a model of my beliefs and intentions	4.90	1.14

tend to support the identification of the partner's negotiation style (mean = 5.38, SD = 1.20) and the learning of different negotiation styles (mean = 5.48, SD = 1.32). The AI-models behaving realistic is rated as neutral by the participants (mean = 4.10, SD = 1.55). The participants tend to agree that the AI models seem to work with a model of their beliefs and intentions (mean = 4.90, SD = 1.14). 21 of the participants completed the survey (9 females; 12 males), 16 of whom were between 18 and 24 years old; two were between 25–34 years old; one was between 45–54 years old; and two were between 55–64 years old.

7 Concluding Discussion

Since the sample size of the conducted experiment is not very large, further investigations have to be conducted. However, the results already show some first implications.

The negotiation styles were correctly identified by 14 participations (26.92%) which is higher than the expected 20% of unsubstantiated decisions. However, their identification varies highly between the styles; Styles in which the agents do not focus on the relationship at all (namely competing and avoiding), were identified correctly in a higher number than the remaining styles; competing with 54.55% (6 of 11), avoiding with 41.67% (5 of 12), accommodating with 27.27% (3 out 11), compromising with 0%, and collaborating with 0%. Since the agents currently only consider the utility values, it can be more challenging to identify them correctly as the utility value is the only parameter to evaluate whether the agents are concerned with the relationship or not. The used preference models in the negotiations are indicated to be collaborative, i.e., the agent could fulfil its behaviour accurately by making concessions in terms of the relationship, but those concession could have a negative effect on the participant's view. Thus, further investigations have to be conducted to evaluate whether the compromising and collaborating agents behave according to the style definitions, e.g., with a larger number of sample size and/or with textual offer exchanges to underline the relationship aspect.

Further, participants with at least some experience were able to identify the models with 40.00% accuracy in general and, thus, performed far better than participants without experience, who only identified correctly with a rate of 18.75%. Thus, negotiation experience supports the correct identification of the partner's style.

The results of the post-negotiation survey (see Table 3) indicate that the approach taken by this work in teaching negotiation styles is well received both as a tool to learn about negotiation styles as well as in identifying them. Thus, the requirement *realistic* is fulfilled. Further, the participants tend to agree with the sentiment that the agents work with a model of their beliefs and intentions, i.e., perform *ToM*. The participants rated the AI-models realistic behaviour as neutral which can be caused by not providing explanations to the offers. The results of Table 3 correspond to the first two derived requirements (cf. Sect. 4.1). The requirement *giving feedback* was fulfilled by revealing the imitated negotiation style after participants gave their judgements, allowing them to reflect on their identification. The agents *shared information* implicitly via the offers values as they are non-textual. The remaining requirements (*no knowledge about the partner's utility* and *use history of exchange*) are met by the design of the agents.

There are some limitations of this research contribution. The systematic literature review is limited by the search string and the data. As already mentioned, the sample size is currently small and further experiments have to be conducted. However, the first implications indicate that the designed agents work according to their purpose and mostly imitate their negotiation styles correctly for the decision-making process. This research contribution demonstrates that no extensive datasets are needed to build agents which can negotiate with specific negotiation styles rather than genetic algorithms can be implemented and trained to do so. However, it is important to note that the agents' current focus is on decision-making. In future research, it would be beneficial to consider the influence of additional factors on real-world negotiations, such as communication and emotions. These factors warrant further examination to enhance our understanding of the complex dynamics involved in negotiations.

As a next step, the agents should be validated in a further experiment by increasing the sample size and conducting a statistical test. Depending on the results, the agents' negotiation styles could be refined though alterations to the genetic algorithm and the incentives (fitness and corresponding weights) or could be compared to different AI methods.

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